3.] ISLR 12.6.2

a.] Step 1: 4 singleton cluster: {A}, {B}, {C}, {D}

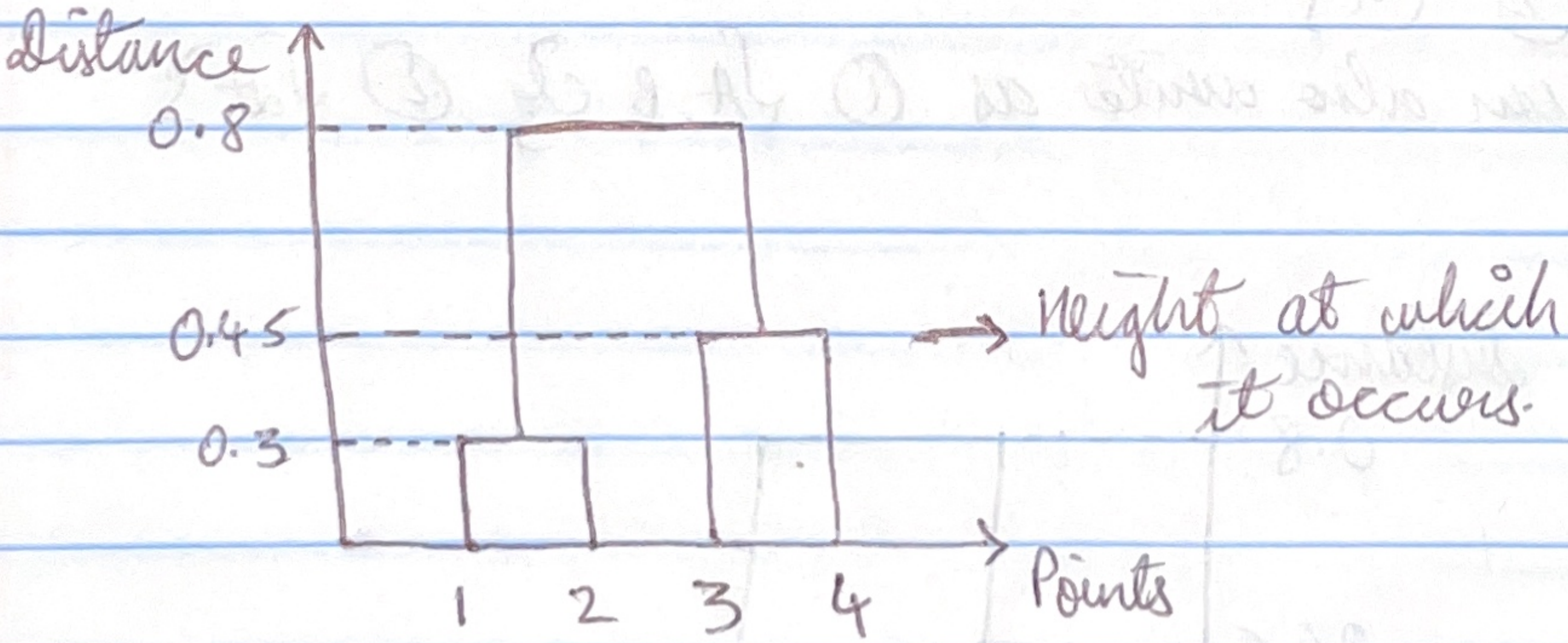
Step 2: Find smallest dissimilarity

A & B : 0.3

so, {A B}, {C}, {D}

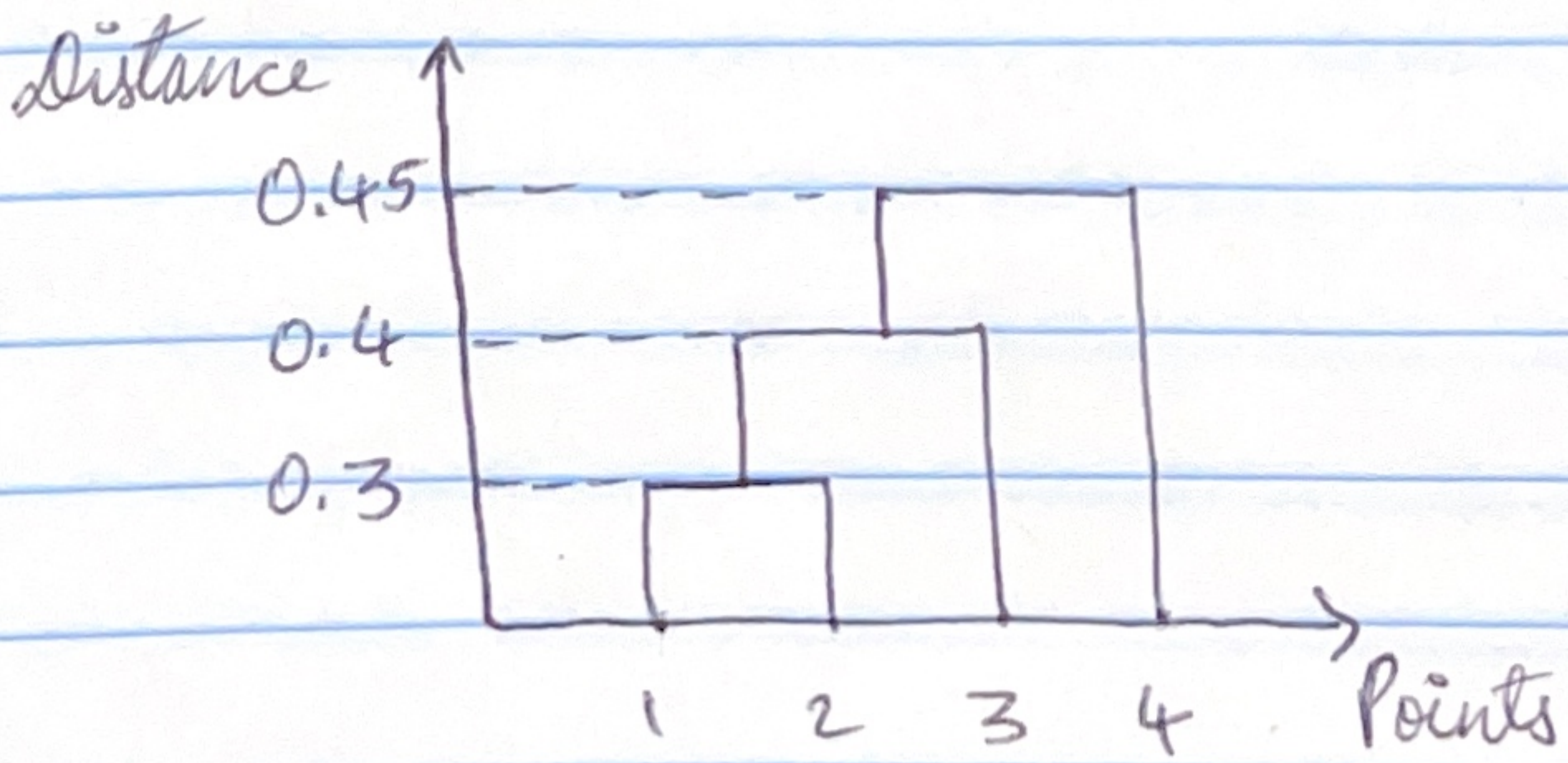
Step 3: Compute dissimilarity from AB to other cluster: smallest → C, D : merge: {C, D} (0.45)

Step 4: d (AB, CD) = max (0.4, 0.7, 0.5, 0.8) = 0.8



Complete linkage Dendogram

b.] Single linkage Dendogram:



C] Observations in the clusters

For cut complete linkage → (2 cluster)

① {1, 2}

② {3, 4}

can also write as ① {A, B}, ② {C, D}

D] Observations in the cluster

cut single linkage dendogram into 2 cluster :

① {1, 2, 3}

② {4}

can also write as ① {A, B, C}, ② {D}

E]